

✓ What is a Functional Interface in Java 1.8?

A **Functional Interface** in Java is an interface that contains **exactly one abstract method**.

Java 1.8 introduced **Lambda Expressions**, and functional interfaces are required to use lambdas.

□ In simple words:

If an interface has **only one abstract method**, it is called a *functional interface*.

□ Why Was Functional Interface Introduced in Java 1.8?

In **Java 1.8**, Oracle introduced:

- Lambda Expressions
- Method References
- Stream API

To support **lambda expressions**, Java needed a target type — and that target type is a **functional interface**.

Lambda expressions can only be used where an interface has **one abstract method**.

□ Definition

A functional interface:

- Has **only one abstract method**
 - Can have:
 - Multiple default methods
 - Multiple static methods
 - Can be annotated with `@FunctionalInterface` (optional but recommended)
-

□ Example of Functional Interface

```
@FunctionalInterface
interface MyFunctionalInterface {
    void sayHello();                // Only one abstract method
}
```

❑ Using Functional Interface with Lambda Expression

```
public class Main {  
    public static void main(String[] args) {  
  
        MyFunctionalInterface obj = () -> {  
            System.out.println("Hello from Lambda!");  
        };  
  
        obj.sayHello();  
    }  
}
```

✓ Output:

Hello from Lambda!

Here:

- `() -> {}` is the **lambda expression**
- It implements the `sayHello()` method

❑ Why Is It Called “Functional”?

Because it represents a **single function behavior**.

It focuses on “what to do” rather than “how to create a class”.

Before Java 1.8, we had to write:

```
MyFunctionalInterface obj = new MyFunctionalInterface() {  
    @Override  
    public void sayHello() {  
        System.out.println("Hello from Anonymous Class");  
    }  
};
```

Java 1.8 made it shorter using lambda.

❑ Difference Between Functional Interface and Other Interfaces

Feature	Functional Interface	Normal Interface
Abstract Methods	Only 1	One or more
Lambda Support	✓ Yes	✗ No
Annotation	@FunctionalInterface	Not required
Purpose	Behavior / Function	Contract for multiple methods
Example	Runnable, Comparator	List, Map

❑ Example of Normal Interface

```
interface NormalInterface {  
    void method1();  
    void method2();  
}
```

This is **NOT** a functional interface because it has **two abstract methods**.

You cannot use lambda here.

❑ Built-in Functional Interfaces in Java 8

Java 8 provides many built-in functional interfaces in:

❑ java.util.function package

Interface	Abstract Method	Purpose
Runnable	run()	Thread execution
Callable	call()	Returns value
Comparator	compare()	Compare objects
Predicate<T>	test(T t)	Returns boolean
Function<T,R>	apply(T t)	Takes input, returns output
Consumer<T>	accept(T t)	Takes input, returns nothing

Supplier<T>	get()	Returns value

□ Example Using Built-in Functional Interface

```
import java.util.function.Predicate;

public class Main {
    public static void main(String[] args) {

        Predicate<Integer> isEven = (num) -> num % 2 == 0;

        System.out.println(isEven.test(10));           // true
        System.out.println(isEven.test(7));           // false
    }
}
```

□ Important Rules

A functional interface:

- ✓ Must have exactly **one abstract method**
- ✓ Can have multiple default methods
- ✓ Can have multiple static methods
- ✓ Can override Object class methods

Example Showing Default + Static Methods

```
@FunctionalInterface
interface Calculator {
    int add(int a, int b);           // Only one abstract method

    default void print() {
        System.out.println("Calculator Interface");
    }

    static void info() {
        System.out.println("Static method inside interface");
    }
}
```

This is still a functional interface because there is only **one abstract method**.

□ Summary

Point	Explanation
What is it?	Interface with exactly one abstract method
Introduced in	Java 8
Why?	To support Lambda expressions
Annotation	@FunctionalInterface (optional)
Main Advantage	Cleaner and shorter code

□ Interview Answer (Short Version)

A functional interface in Java 8 is an interface that contains exactly one abstract method. It enables lambda expressions and method references. Unlike normal interfaces, which can have multiple abstract methods, a functional interface focuses on a single behavior. It may contain multiple default and static methods but only one abstract method.

If you want, I can also explain:

- Functional Interface vs Anonymous Class (deep comparison)
- Real-time industry example
- How Streams use functional interfaces
- Java 8 internal working mechanism

▣ 1 Functional Interface vs Anonymous Class (Deep Comparison)

Before Java 8, we mainly used **Anonymous Inner Classes**.

After Java 8, **Lambda Expressions + Functional Interfaces** became preferred.

□ Basic Concept Difference

Feature	Functional Interface + Lambda	Anonymous Class
Code Length	Very Short	Long & Verbose

Readability	Clean	More boilerplate
Performance	Slightly better (invokedynamic)	Creates separate class
Syntax	() -> {}	new Interface() {}
Best For	Single behavior	Multiple methods / complex logic
Access to this	Refers to outer class	Refers to anonymous class

❑ Example Comparison

✔ Using Anonymous Class (Before Java 8)

```

interface Greeting {
    void sayHello();
}

public class Main {
    public static void main(String[] args) {

        Greeting obj = new Greeting() {
            @Override
            public void sayHello() {
                System.out.println("Hello from Anonymous Class");
            }
        };

        obj.sayHello();
    }
}

```

✔ Using Functional Interface + Lambda (Java 8)

```

@FunctionalInterface
interface Greeting {
    void sayHello();
}

public class Main {
    public static void main(String[] args) {

        Greeting obj = () ->
            System.out.println("Hello from Lambda");
    }
}

```

```
    obj.sayHello();  
  }  
}
```

☐ Key Internal Difference

Anonymous Class:

- Compiler creates a **separate .class file**
- Uses traditional object creation

Lambda:

- Uses **invokedynamic**
 - No extra class file
 - More optimized at runtime
-

☐ When to Use What?

Scenario	Recommended
Single method behavior	Lambda
Multiple abstract methods	Anonymous class
Complex logic inside	Anonymous class
Simple callback	Lambda

☐ 2 Real-Time Industry Examples

Let's take real-world scenarios.

☐ Example 1: Sorting Employees (Comparator)

In enterprise applications (HR systems, payroll systems), sorting is very common.

Without Lambda (Old Way)

```
Collections.sort(employeeList, new Comparator<Employee>() {  
    public int compare(Employee e1, Employee e2) {
```

```
    return e1.getSalary() - e2.getSalary();  
    }  
});
```

With Lambda (Modern Way)

```
employeeList.sort((e1, e2) -> e1.getSalary() - e2.getSalary());
```

✓ Used in:

- Banking systems
 - Payroll systems
 - E-commerce product sorting
-

❑ Example 2: Runnable in Multithreading

Old Way

```
Thread t = new Thread(new Runnable() {  
    public void run() {  
        System.out.println("Thread running");  
    }  
});
```

Java 8 Lambda Way

```
Thread t = new Thread(() ->  
    System.out.println("Thread running"));  
t.start();
```

✓ Used in:

- Microservices
 - Backend APIs
 - Background job processing
-

❑ 3 How Streams Use Functional Interfaces

Java 8 Streams are heavily based on **functional interfaces**.

Streams use:

Stream Method	Functional Interface Used
filter()	Predicate
map()	Function
forEach()	Consumer
reduce()	BinaryOperator
sorted()	Comparator

❑ Practical Example Using Stream

```
import java.util.*;
import java.util.stream.*;

public class Main {
    public static void main(String[] args) {

        List<Integer> numbers = Arrays.asList(10, 15, 20, 25, 30);

        numbers.stream()
            .filter(n -> n % 2 == 0)                // Predicate
            .map(n -> n * 2)                        // Function
            .forEach(n -> System.out.println(n));    // Consumer
    }
}
```

❑ What Happens Internally?

- filter() accepts a **Predicate**
- map() accepts a **Function**
- forEach() accepts a **Consumer**

Each lambda you write is actually implementing a functional interface.

❑ Real Enterprise Stream Example

Imagine an e-commerce system:

```
List<Product> expensiveProducts =
    products.stream()
        .filter(p -> p.getPrice() > 5000)
```

```
.sorted((p1, p2) -> p2.getPrice() - p1.getPrice())  
.collect(Collectors.toList());
```

Used in:

- Flipkart/Amazon-style filtering
 - Banking transaction filtering
 - Analytics dashboards
-

❑ Why Streams + Functional Interfaces Are Powerful?

- ✓ Less boilerplate
 - ✓ More readable
 - ✓ Parallel processing support
 - ✓ Declarative programming style
 - ✓ Reduced bugs
-

❑ Interview-Level Summary

Functional Interface vs Anonymous Class:

- Lambda is shorter, faster, and cleaner.
- Anonymous class creates a new class file.
- Lambda uses invokedynamic.

Real-Time Usage:

- Sorting (Comparator)
- Filtering (Predicate)
- Multithreading (Runnable)
- Event handling

Streams:

Streams depend entirely on functional interfaces like:

- Predicate
- Function
- Consumer
- Supplier
- Comparator

If you want, I can next explain:

- Internal working of Lambda (Bytecode level)
- Functional Interface memory model
- Method Reference vs Lambda
- Complete Stream lifecycle diagram ☐

Excellent ☑

These five are the **core functional interfaces** in `java.util.function` and are heavily used in **Streams, collections, multithreading, and enterprise applications**.

I'll explain each with:

- ✓ Definition
- ✓ Method Signature
- ✓ When to Use
- ✓ Simple Java Example
- ✓ Real Industry Example

1 Predicate

✓ Definition

A **Predicate** is a functional interface that:

Takes one input and returns a boolean result.

Used mainly for **filtering and condition checking**.

☐ Package: `java.util.function.Predicate`

☐ Abstract Method

```
boolean test(T t);
```

☐ Simple Example

```
import java.util.function.Predicate;

public class Main {
    public static void main(String[] args) {
```

```
Predicate<Integer> isEven = num -> num % 2 == 0;

System.out.println(isEven.test(10));           // true
System.out.println(isEven.test(7));           // false
}
}
```

❑ Real Industry Example (Filtering Users)

```
List<String> users = Arrays.asList("Anurag", "Admin", "Guest");

Predicate<String> isAdmin = name -> name.equals("Admin");

users.stream()
    .filter(isAdmin)
    .forEach(System.out::println);
```

✓ Used in:

- Authentication systems
 - Transaction validation
 - Fraud detection
-

2 Function<T, R>

✓ Definition

A Function:

Takes one input and returns one output.

Used for **data transformation**.

❑ Package: java.util.function.Function

❑ Abstract Method

```
R apply(T t);
```

❑ Simple Example

```
import java.util.function.Function;

public class Main {
    public static void main(String[] args) {

        Function<Integer, String> convertToString =
            num -> "Number is: " + num;

        System.out.println(convertToString.apply(5));
    }
}
```

☐ **Real Industry Example (Salary Calculation)**

```
Function<Double, Double> calculateBonus =
    salary -> salary * 1.10;

System.out.println(calculateBonus.apply(50000.0));
```

✓ Used in:

- Data transformation
 - API response mapping
 - DTO conversion
-

3 Consumer

✓ **Definition**

A Consumer:

Takes one input and returns nothing (void).

Used for **performing actions**.

☐ Package: java.util.function.Consumer

☐ **Abstract Method**

```
void accept(T t);
```

☐ **Simple Example**

```
import java.util.function.Consumer;

public class Main {
    public static void main(String[] args) {

        Consumer<String> printName =
            name -> System.out.println("Hello " + name);

        printName.accept("Anurag");
    }
}
```

☐ Real Industry Example (Logging)

```
List<String> logs = Arrays.asList("Login", "Payment", "Logout");

logs.forEach(log -> System.out.println("Event: " + log));
```

✓ Used in:

- Logging systems
 - Sending notifications
 - Writing to database
-

4 Supplier

✓ Definition

A Supplier:

Takes no input but returns a value.

Used for **generating or supplying data**.

☐ Package: java.util.function.Supplier

☐ Abstract Method

T get();

☐ Simple Example

```
import java.util.function.Supplier;

public class Main {
    public static void main(String[] args) {

        Supplier<Double> randomValue =
            () -> Math.random();

        System.out.println(randomValue.get());
    }
}
```

❑ Real Industry Example (OTP Generator)

```
Supplier<Integer> otpGenerator =
    () -> (int)(Math.random() * 10000);

System.out.println("OTP: " + otpGenerator.get());
```

✓ Used in:

- OTP systems
 - Lazy initialization
 - Object creation factories
-

5 Comparator

✓ Definition

A Comparator:

Compares two objects and returns an integer.

❑ Package: java.util.Comparator

❑ Abstract Method

```
int compare(T o1, T o2);
```

Return value meaning:

Return	Meaning
--------	---------

0	Equal
> 0	First greater
< 0	Second greater

❑ Simple Example

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        List<Integer> numbers = Arrays.asList(5, 2, 8, 1);

        numbers.sort((a, b) -> a - b);

        System.out.println(numbers);
    }
}
```

❑ Real Industry Example (Sort Employees by Salary)

```
class Employee {
    String name;
    int salary;

    Employee(String name, int salary) {
        this.name = name;
        this.salary = salary;
    }
}

List<Employee> list = Arrays.asList(
    new Employee("A", 50000),
    new Employee("B", 70000)
);

list.sort((e1, e2) -> e2.salary - e1.salary);
```

✓ Used in:

- E-commerce sorting
- HR payroll system
- Banking transaction ranking

□ Summary Table

Interface	Input	Output	Purpose
Predicate	1	boolean	Condition check
Function	1	1	Transform data
Consumer	1	void	Perform action
Supplier	0	1	Provide data
Comparator	2	int	Compare objects

□ How Streams Use Them Together

```
List<Integer> list = Arrays.asList(10, 15, 20, 25);
```

```
list.stream()  
    .filter(n -> n > 15)           // Predicate  
    .map(n -> n * 2)              // Function  
    .forEach(System.out::println); // Consumer
```

□ Interview Summary (Strong Answer)

- Predicate → condition checking
- Function → transformation
- Consumer → action performer
- Supplier → value provider
- Comparator → object comparison

All are **functional interfaces** introduced in Java 8 and form the backbone of the Stream API.

If you want next:

- Memory diagram explanation
- Internal working (bytecode level)
- Real microservice-level example
- Complete CRUD project example using all five

Just tell me □

Excellent question ☺ — this goes deeper into **how lambda works internally in Java 8**.

Let's break it clearly.

1 What is a "Target Type" in Java?

✔ Definition

A **Target Type** is:

The data type (usually a functional interface) that the compiler uses to determine the type of a lambda expression.

In simple words:

- ❑ A lambda expression does NOT have its own type.
- ❑ It gets its type from the context where it is used.

That context type is called the **Target Type**.

❑ Why Does Java Need a Target Type?

Because this is **NOT valid in Java**:

```
() -> System.out.println("Hello");
```

Java doesn't know:

- What method is this implementing?
- What is the return type?
- What are the parameters?

So Java needs a **functional interface reference** to assign the lambda.

❑ Example Without Target Type (✗ Error)

```
() -> System.out.println("Hello"); // Compile-time error
```

❑ Example With Target Type (✔ Correct)

```
@FunctionalInterface
```

```

interface Greeting {
    void sayHello();
}

public class Main {
    public static void main(String[] args) {

        Greeting g = () -> System.out.println("Hello");

        g.sayHello();
    }
}

```

Here:

Greeting is the **Target Type**

The compiler understands:

- Greeting has one method → void sayHello()
- So lambda must match that method

□ Another Example (Very Important)

```

import java.util.function.Predicate;

public class Main {
    public static void main(String[] args) {

        Predicate<Integer> isEven = n -> n % 2 == 0;

        System.out.println(isEven.test(10));
    }
}

```

Target Type = Predicate<Integer>

The compiler sees:

```
boolean test(Integer n);
```

So it checks:

- Does lambda return boolean? ✓
- Does it take one Integer parameter? ✓

So it compiles successfully.

☐ Target Type Can Come From:

Context	Example
Variable assignment	Predicate<Integer> p = n -> n > 5;
Method parameter	list.sort((a,b) -> a-b);
Return type	return x -> x*x;
Type casting	(Runnable) () -> System.out.println("Hi");

2 What is “Single-Function Behavior”?

✔ Definition

Single-function behavior means:

The interface represents exactly ONE behavior (one action / one task).

A functional interface defines:

- ☐ One abstract method
 - ☐ One behavior
 - ☐ One logical operation
-

☐ Example 1 – Runnable

Runnable r = () -> System.out.println("Running task");

What is the behavior?

✔ "Run something"

Only one job: run()

☐ Example 2 – Predicate

```
Predicate<Integer> isAdult = age -> age >= 18;
```

Single behavior?

✓ "Check condition"

Only one job: test()

□ **Example 3 – Function**

```
Function<String, Integer> getLength = str -> str.length();
```

Single behavior?

✓ "Transform input to output"

Only one job: apply()

□ **Why Java Requires Single-Function Behavior?**

Because lambda represents:

- A single block of logic
- A single implementation

If an interface had multiple abstract methods:

```
interface Example {  
    void method1();  
    void method2();  
}
```

Then this would be confusing:

Example e = () -> ??? // Which method?

Java wouldn't know which method the lambda is implementing.

That's why:

- ✓ Functional Interface = Exactly 1 abstract method
- ✓ So lambda maps clearly to that one method

❑ Real-World Analogy

Think of it like this:

Functional Interface = One Button Machine

You press button → One action happens.

Normal Interface = Multiple Button Machine

You press button → Many options possible.

Lambda works only when there is ONE button.

❑ Deep Technical Understanding

Lambda expression internally:

- Does NOT create a new anonymous class
 - Uses invokedynamic
 - Binds dynamically to a functional interface
 - Needs a target type for type inference
-

❑ Final Summary

✔ Target Type

- The functional interface type that provides context to a lambda expression.
 - Without it, lambda cannot compile.
 - Comes from assignment, method parameter, or return type.
-

✔ Single-Function Behavior

- Means one abstract method.
 - Represents one logical operation.
 - Allows lambda to map clearly to that method.
-

❑ Interview-Level Answer

Target type is the functional interface type that provides context to a lambda expression. Since lambda expressions do not have their own type, the compiler uses the target type to determine which abstract method the lambda implements.

Single-function behavior means the interface defines exactly one abstract method, representing one logical operation, which allows lambda expressions to implement it unambiguously.

If you want next:

- Type inference deep explanation
- invokedynamic internal JVM working
- Memory model difference between lambda and anonymous class
- Bytecode comparison

✓ Meaning of “Single Function Behavior” in Java

□ Simple Definition

Single-function behavior means an interface represents **only one logical operation (one behavior)** through **exactly one abstract method**.

□ Break It Down

- **Single** → Only one
- **Function** → Method (operation)
- **Behavior** → What the method does

□ So:

A functional interface defines **one action / one responsibility / one behavior**

□ Example 1: Runnable

```
Runnable r = () -> System.out.println("Task is running");
```

What is the behavior?

✓ “Run a task”

Only one method exists:

```
void run();
```

- ☐ This interface represents **one behavior only** → **running a task**
-

☐ **Example 2: Predicate**

```
Predicate<Integer> isEven = n -> n % 2 == 0;
```

Behavior?

✓“Check condition”

Only one method:

```
boolean test(T t);
```

- ☐ One behavior → return true/false
-

☐ **Example 3: Function**

```
Function<String, Integer> length = s -> s.length();
```

Behavior?

✓“Transform input to output”

Only one method:

```
R apply(T t);
```

- ☐ One behavior → conversion
-

✗ What is NOT Single-Function Behavior?

```
interface Example {  
    void method1();  
    void method2();  
}
```

☐ This has **multiple behaviors**:

- method1 → behavior 1
- method2 → behavior 2

So this is **NOT a functional interface**

☐ **Why Single-Function Behavior is Important**

Because of **Lambda Expressions**

Example `e = () -> ???` // **✗**Confusion

☐ Which method should lambda implement?

Java cannot decide.

✓ **With Single Behavior (Works)**

```
interface Greeting {  
    void sayHello();  
}
```

`Greeting g = () -> System.out.println("Hello");`

✓ No confusion

✓ Lambda maps directly to one method

☐ **Real-World Analogy**

☐ **Functional Interface (Single Behavior)**

Think of a **remote with only ONE button**

☐ Press → One action happens

☐ **Normal Interface (Multiple Behaviors)**

Think of a **remote with MANY buttons**

☐ Which button does lambda represent? ✕

☐ **Key Insight**

Lambda = implementation of **one method only**

So Java requires:

- ✓ One abstract method
 - ✓ One behavior
 - ✓ One clear mapping
-

☐ **Interview Answer**

Single-function behavior means that a functional interface represents exactly one logical operation through a single abstract method. This allows lambda expressions to provide a direct and unambiguous implementation of that method.

If you want, I can go deeper into:

- How JVM maps lambda to that single method
- Bytecode-level explanation
- Real Spring Boot examples using this concept ☐

Target Type in Java

The context where a lambda expression or method reference is used determines the **target type**.

Assigning to a Functional Interface:

```
Runnable r = () →  
    System.out.println("Hello");
```

Target Type: **Runnable**

Passing as a Method Argument:

```
list.forEach(s → )  
    System.out.println(s);
```

Target Type: **Consumer<String>**

???

Return from a Method:

```
return (a, b) → a + b;
```

Target Type: **BiFunction<Integer, Integer, Integer>**

Used in a Conditional Expression:

```
Predicate<String> p = flag ?  
    s → s.isEmpty() : String::isBlank;
```

Target Type: **Predicate<String>**

The target type is the expected type that the lambda expression or method reference is compatible with.

It determines **how the lambda** is interpreted.



✓ Meaning of Target Type in Java

In Java, **target type** means:

- **The expected type of an expression based on the context in which it is used.**
-

□ Simple Explanation

When you write code (especially **lambda expressions** or **method references**), Java does not always know the type directly.

So it looks at the **context (where the expression is used)** to determine the type — this is called the **target type**.

□ Key Idea

- Java is **context-aware**
 - The **surrounding code decides the type**
 - Mostly used in:
 - Lambda expressions
 - Method references
 - Generics
-

□ Examples

1. Lambda with Assignment

```
Runnable r = () -> System.out.println("Hello");
```

- Target type = Runnable
Because the lambda is assigned to a Runnable variable.
-

2. Lambda as Method Argument

```
list.forEach(s -> System.out.println(s));
```

- Target type = Consumer<String>
Because `forEach()` expects a Consumer.
-

3. Return from Method

`return (a, b) -> a + b;`

□ Target type = `BiFunction<Integer, Integer, Integer>`
Because the method return type defines it.

4. Conditional Expression

`Predicate<String> p = flag ? s -> s.isEmpty() : String::isBlank;`

□ Target type = `Predicate<String>`

⚡ One-Line Definition

Target type = the type that Java expects in a given context, used to interpret expressions like lambdas.

□ Why It Matters

- Helps Java **infer types automatically**
 - Reduces need for **explicit type declarations**
 - Makes code **cleaner and shorter**
-

If you want, I can give you **interview questions + tricky examples** on target types (very useful for Java developer roles).